



ISO 9001: 2015
SOCOTEC SCP000722Q
REPUBLIC OF THE



PHILIPPINES **BICOLUNIVERSITY POLANGUI**
Polangui, Albay

Name: Kehn Rebadavia

Subject: System Administration and Maintenance

Course, Year & Section: BSIT 4B

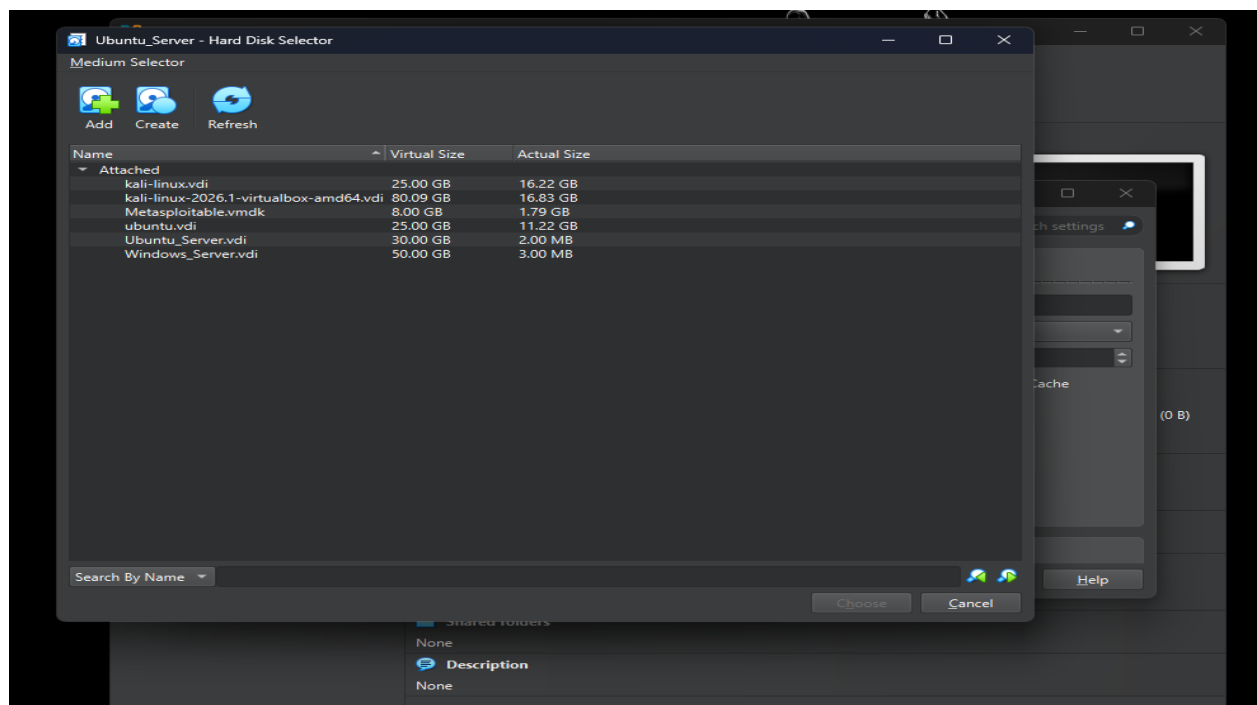
Prof: Guillermo V. Red

Laboratory Guide for Week 4

Create a Virtual Disk

A. Partition & Mount

- Device Name: /dev/sdb1
- File System Type: ext4
- Mount Point: /projectdata
- Screenshot(s)/Command Outputs:



Detect New Disk

- Identify as /dev/sdb (or similar).

ubuntu.vdi	25.00 GB	11.22 GB
Ubuntu_Server.vdi	30.00 GB	2.00 MB
Windows_Server.vdi	50.00 GB	3.00 MB
▼ Not Attached		
Ubuntu_Server_1.vdi	10.00 GB	2.00 MB

Partition and Format

```
an error occurred. Press enter to start a shell

root@ubuntu-server:/var/snap/subiquity/7225#
root@ubuntu-server:/var/snap/subiquity/7225#
root@ubuntu-server:/var/snap/subiquity/7225#
root@ubuntu-server:/var/snap/subiquity/7225# lsblk
NAME MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0  7:0    0   1.1G  1 loop /rofs
loop1  7:1    0 148.2M  1 loop
loop2  7:2    0 141.3M  1 loop
loop3  7:3    0   96K   1 loop
loop4  7:4    0  48.4M  1 loop /snap/snapd/26382
loop5  7:5    0  21.7M  1 loop /snap/subiquity/7225
loop6  7:6    0  66.8M  1 loop /snap/core24/1587
sda     8:0    0   30G   0 disk
sdb     8:16   0   10G   0 disk
sr0     11:0   1   2.7G   0 rom  /cdrom
root@ubuntu-server:/var/snap/subiquity/7225# _
```

- Create a new partition.
- Write changes.
- Format with ext4:

```
Welcome to fdisk (util-linux 2.41.3).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS (MBR) disklabel with disk identifier 0x7d9ce6e2.

Command (m for help): n
Partition type
  p   primary (0 primary, 0 extended, 4 free)
  e   extended (container for logical partitions)
Select (default p): p
Partition number (1-4, default 1): 1
First sector (2048-20971519, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-20971519, default 20971519):

Created a new partition 1 of type 'Linux' and of size 10 GiB.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

root@ubuntu-server:/var/snap/subiquity/7225#
```

B. Directory Setup & Permissions

- Directory Path: /projectdata
- Group Assigned: root:facultygrp (with studentgrp restricted via others)
- Permission Level: Faculty (facultygrp = Read/Write), Students (studentgrp / Others = Read-only via 775)
- Screenshot(s)/Command Outputs:

```
sudo mkdir /mnt/data sudo mount /dev/sdb1 /mnt/data
```

Assign Group Permissions

```
root@ubuntu-server:/var/snap/subiquity/7225# chown root:facultygrp /projectdata
chown: invalid group: 'root:facultygrp'
root@ubuntu-server:/var/snap/subiquity/7225# groupadd facultygrp
root@ubuntu-server:/var/snap/subiquity/7225# groupadd studentgrp
root@ubuntu-server:/var/snap/subiquity/7225# useradd -m -g facultygrp faculty2
useradd: group 'facultygrp' does not exist
root@ubuntu-server:/var/snap/subiquity/7225# groupadd facultygrp
root@ubuntu-server:/var/snap/subiquity/7225# useradd -m -g facultygrp faculty2
root@ubuntu-server:/var/snap/subiquity/7225# useradd -m -g studentgrp student4
root@ubuntu-server:/var/snap/subiquity/7225# chown root:facultygrp /projectdata
chown: cannot dereference 'facultygrp': No such file or directory
root@ubuntu-server:/var/snap/subiquity/7225# chown root:facultygrp /projectdata
root@ubuntu-server:/var/snap/subiquity/7225# chmod 770/projectdata
chmod: missing operand
Try 'chmod --help' for more information.
root@ubuntu-server:/var/snap/subiquity/7225# chmod 770 /projectdata
root@ubuntu-server:/var/snap/subiquity/7225#
```

Verification & Troubleshooting

- Log in as faculty2 → Should be able to write inside /mnt/data.

```
su: user faculty2 does not exist or the user entry does not contain all the req
root@ubuntu-server:/var/snap/subiquity/7225# useradd -m -g facultygrp faculty2
root@ubuntu-server:/var/snap/subiquity/7225# chmod 770 /projectdata
root@ubuntu-server:/var/snap/subiquity/7225# su - faculty2
$ cd /projectdata
$ touch test.txt
$ exit
root@ubuntu-server:/var/snap/subiquity/7225# _
```

- Log in as student4 → Should be denied write access.

```
root@ubuntu-server:/var/snap/subiquity/7225# chmod 775 /projectdata
root@ubuntu-server:/var/snap/subiquity/7225# su - student4
$ cd /projectdata
$ ls
lost+found  test.txt
$ touch test.txt
touch: setting times of 'test.txt': Permission denied
$ exit
root@ubuntu-server:/var/snap/subiquity/7225#
```

Part 3 – Reflection

1. What challenges did you encounter in creating partitions and managing permissions?

Ensuring the correct identification of storage devices via lsblk was vital to prevent altering core system partitions. Additionally, configuring multi-user access constraints required balancing directory permissions (chmod 775) so that faculty members retained full read/write privileges while students were safely restricted to read-only access.

2. How can these skills be applied in real-world system administration duties?

These competencies are critical for scaling server infrastructure dynamically when storage demands increase without interrupting core services. Managing filesystems and enforcing strict access control lists ensures that enterprise data remains secure, organized, and compliant with corporate roles.